WORKED SOLUTION

STEP III 2022, Question 7

Pure • Vector Product Transformations

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.5, *Rotating in Three Dimensions*: https://beyondstep.keithmcnulty.org/ch07/rotations-3d.html

The key idea

Everything rests on splitting a vector into a part along $\mathbf{n}$ and a part perpendicular to $\mathbf{n}$. The map $\mathbf{r} \mapsto \mathbf{n} \times \mathbf{r}$ kills the parallel part and turns the perpendicular part through a right angle about $\mathbf{n}$ without changing its length. Once part (i) shows that applying it twice gives minus the perpendicular part, the maps in (ii) and (iii) can be read off by seeing what they do to each of the two parts separately.

Solution

Part (i)

With $\mathbf{n} = (a, b, c)^{\mathsf{T}}$ and $\mathbf{r} = (x, y, z)^{\mathsf{T}}$,

$$\mathrm{f}(\mathbf{r}) = \mathbf{n} \times \mathbf{r} = \begin{pmatrix} bz - cy \\ cx - az \\ ay - bx \end{pmatrix}.$$

Applying f again, the x-component of $\mathbf{n} \times \mathrm{f}(\mathbf{r})$ is b times the third component of $\mathrm{f}(\mathbf{r})$ minus c times the second:

$$\begin{aligned} b(ay - bx) - c(cx - az) &= aby - b^2x - c^2x + acz \\ &= -x(b^2 + c^2) + aby + acz, \end{aligned}$$

as required. Since $\mathbf{n}$ is a unit vector, $b^2 + c^2 = 1 - a^2$, so this becomes

$$-x + a^2x + aby + acz = a(ax + by + cz) - x = (\mathbf{n} \cdot \mathbf{r})\, a - x.$$

That is the x-component of $(\mathbf{n} \cdot \mathbf{r})\mathbf{n} - \mathbf{r}$. The y- and z-components follow in the same way by cycling $a \to b \to c \to a$ and $x \to y \to z \to x$, since the cross product is

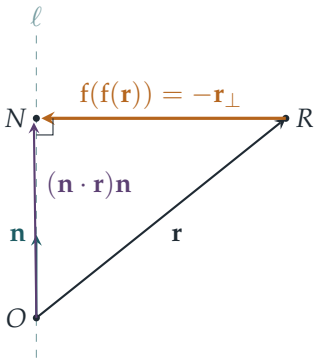
unchanged by this cyclic relabelling. Hence

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = (\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r}.$$

Geometric meaning. Write $\mathbf{r} = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp}$, where $\mathbf{r}_{\parallel} = (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$ and $\mathbf{r}_{\perp} = \mathbf{r} - (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is perpendicular to $\mathbf{n}$. Then

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}_{\perp}.$$

Let ℓ be the line through O in the direction of $\mathbf{n}$ and let N be the foot of the perpendicular from R to ℓ, so $\overrightarrow{ON} = \mathbf{r}_{\parallel}$. Then $\mathrm{f}(\mathrm{f}(\mathbf{r})) = \overrightarrow{ON} - \overrightarrow{OR} = \overrightarrow{RN}$: it is the vector from R straight to the axis ℓ, at right angles to it. Its length is the distance of R from ℓ.



Part (ii)

The axis. Since $\mathbf{n}\times\mathbf{n} = \mathbf{0}$, we have $\mathrm{f}(\mathbf{n}) = \mathbf{0}$ and so $\mathrm{f}(\mathrm{f}(\mathbf{n})) = \mathbf{0}$, giving $\mathrm{g}(\mathbf{n}) = \mathbf{n}$. By linearity $\mathrm{g}(\lambda\mathbf{n}) = \lambda\mathbf{n}$, so every point of the line ℓ is fixed.

Perpendicular vectors. Now take $\mathbf{r}$ with $\mathbf{n}\cdot\mathbf{r} = 0$. By part (i), $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$, so

$$\mathrm{g}(\mathbf{r}) = \mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r}) - (1-\cos\theta)\mathbf{r} = \cos\theta\,\mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r}).$$

? **Ask yourself** What does $\mathbf{n}\times\mathbf{r}$ look like when $\mathbf{r}\perp\mathbf{n}$?

It is perpendicular to both $\mathbf{n}$ and $\mathbf{r}$, so it lies in the plane through O perpendicular to $\mathbf{n}$, at right angles to $\mathbf{r}$. Its length is $|\mathbf{n}||\mathbf{r}|\sin 90^\circ = |\mathbf{r}|$. So $\mathbf{r}$ and $\mathbf{n}\times\mathbf{r}$ are two perpendicular vectors of equal length in that plane, and $\mathbf{n}\times\mathbf{r}$ is $\mathbf{r}$ turned through 90° in the right-handed sense about $\mathbf{n}$ (anticlockwise when viewed from the tip of $\mathbf{n}$ looking back towards O). Exactly as $\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$ is $\mathbf{i}$ rotated by θ towards $\mathbf{j}$, the vector $\cos\theta\,\mathbf{r} + \sin\theta\,(\mathbf{n}\times\mathbf{r})$ is $\mathbf{r}$ rotated through θ about $\mathbf{n}$ in that same sense.

General vectors. f is linear (the cross product is linear in its second argument), so g is linear too. Writing a general $\mathbf{r}$ as $\mathbf{r}_{\parallel} + \mathbf{r}_{\perp}$, g leaves $\mathbf{r}_{\parallel}$ alone and rotates $\mathbf{r}_{\perp}$ through θ about $\mathbf{n}$, which is exactly what a rotation about ℓ does.

Therefore $R \mapsto P$ is a **rotation through angle θ about the line through O in the direction $\mathbf{n}$**, in the right-handed sense about $\mathbf{n}$ (anticlockwise when looking from the tip of $\mathbf{n}$ towards O).

Part (iii)

Using part (i),

$$\mathrm{h}(\mathbf{r}) = -\mathbf{r} - 2\big((\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r}\big) = \mathbf{r} - 2(\mathbf{n}\cdot\mathbf{r})\mathbf{n}.$$

In terms of the two components, $\mathrm{h}(\mathbf{r}) = \mathbf{r}_\perp - \mathbf{r}_\parallel$:

- $\mathrm{h}(\mathbf{n}) = \mathbf{n} - 2\mathbf{n} = -\mathbf{n}$, so the component along $\mathbf{n}$ is reversed;
- if $\mathbf{n}\cdot\mathbf{r} = 0$ then $\mathrm{h}(\mathbf{r}) = \mathbf{r}$, so every point of the plane $\mathbf{n}\cdot\mathbf{r} = 0$ is fixed.

Geometrically, $\overrightarrow{RQ} = -2(\mathbf{n}\cdot\mathbf{r})\mathbf{n}$ is normal to the plane $\Pi:\ \mathbf{n}\cdot\mathbf{r} = 0$, and the midpoint of RQ has position vector $\mathbf{r} - (\mathbf{n}\cdot\mathbf{r})\mathbf{n}$, which lies in Π. So Q is the mirror image of R: the map $R \mapsto Q$ is **reflection in the plane through O perpendicular to $\mathbf{n}$**.

Checks

- Take $\mathbf{n} = \mathbf{k} = (0,0,1)^{\mathsf{T}}$. Then $\mathbf{n}\times\mathbf{r} = (-y,x,0)^{\mathsf{T}}$, twice gives $(-x,-y,0)^{\mathsf{T}}$, matching $(\mathbf{n}\cdot\mathbf{r})\mathbf{n} - \mathbf{r} = (0,0,z)^{\mathsf{T}} - (x,y,z)^{\mathsf{T}}$. Then $\mathrm{g}(\mathbf{r}) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)^{\mathsf{T}}$, the familiar rotation about the z-axis, and $\mathrm{h}(\mathbf{r}) = (x,y,-z)^{\mathsf{T}}$, reflection in the plane $z = 0$.
- $\theta = 0$ gives $\mathrm{g}(\mathbf{r}) = \mathbf{r}$, and applying h twice gives the identity, as a rotation by nothing and a reflection should.
- A computer check with 2000 random unit vectors $\mathbf{n}$, vectors $\mathbf{r}$ and angles θ confirmed the x-component formula, the identity for $\mathrm{f}(\mathrm{f}(\mathbf{r}))$, that g agrees with a right-handed rotation by θ about $\mathbf{n}$, and that h agrees with reflection in the plane $\mathbf{n}\cdot\mathbf{r} = 0$.

Common slips

- Forgetting to use $a^2 + b^2 + c^2 = 1$: the identity in (i) is false for a non-unit $\mathbf{n}$.
- In (ii), only checking $\mathbf{r} \perp \mathbf{n}$ and not explaining why that, with $\mathrm{g}(\mathbf{n}) = \mathbf{n}$ and linearity, settles every $\mathbf{r}$.
- Calling (iii) a reflection in the *line* ℓ: points on ℓ are reversed, not fixed; it is the plane perpendicular to $\mathbf{n}$ that is fixed.

Beyond STEP

The formula in (ii) is Rodrigues' rotation formula, and the matrix of f is the antisymmetric "cross-product matrix" K, so a rotation is $I + \sin\theta\, K + (1-\cos\theta)K^2$, which is also $\exp(\theta K)$. The map in (iii) is a Householder reflection $I - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}$, a workhorse of numerical linear algebra. In a first-year course these appear as the orthogonal matrices with determinant $+1$ and -1 respectively. Strang's *Introduction to Linear Algebra* is a good place to see them.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.